MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 05 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If α, β are the zeroes of the polynomial $ax^2 - 5x + c$ such that $\alpha + \beta = \alpha\beta = 10$, then: [1]

(A) $a = 5, c = \frac{1}{2}$ (B) $a = 1, c = \frac{5}{2}$ (C) $a = \frac{1}{2}, c = 5$ (D) $a = \frac{1}{2}, c = 10$

Q2. The zeroes of the quadratic polynomial $p(x) = 4x^2 - 12x + 9$ are: [1]

(A) $\frac{3}{2}, \frac{3}{2}$ (B) $-\frac{3}{2}, -\frac{3}{2}$ (C) $3, -3$ (D) $\frac{2}{3}, \frac{2}{3}$

Q3. If α, β are the zeroes of $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, then the value of k is: [1]

(A) 6 (B) 4 (C) 5 (D) 12

Q4. If the zeroes of $x^2 - kx + 6$ are in the ratio $2 : 3$, then the positive value of k is: [1]

(A) 5 (B) -5 (C) 6 (D) 1

Q5. **Assertion (A):** If one zero of $(k^2 + 16)x^2 + 13x + 8k$ is reciprocal of the other, then $k = 4$.
Reason (R): If zeroes of $ax^2 + bx + c$ are reciprocals of each other, then $a = c$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. Form a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. [2]

Q7. If the squared difference of the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ is equal to 144, find the value(s) of p. [3]

Q8. If α and β are the zeroes of the quadratic polynomial $f(x) = 2x^2 - 5x + 7$, find a quadratic polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$. [3]

Q9. Find the zeroes of $f(x) = 4x^2 + 5\sqrt{2}x - 3$ and verify the relationship between the zeroes and the coefficients. [3]

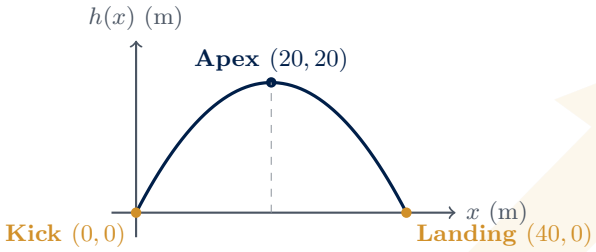
SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — Flight of an Incline Football Kick

During a football match, a striker kicks the ball from ground level ($x = 0$) towards the opponent's goal. The trajectory of the ball in the vertical plane is modelled by the quadratic polynomial:

$$h(x) = -\frac{1}{20}x^2 + 2x$$

where x is the horizontal distance travelled (in metres) and $h(x)$ is the height of the ball above the turf (in metres).



Based on the above projectile model, answer the following questions:

(i) State the geometric shape traced by the football and determine whether the parabola opens upward or downward. [1]

(ii) Find the horizontal distance x where the football touches the turf again. [1]

(iii) **(a)** Determine the maximum height attained by the football during its flight. [2]

— OR —

(b) The goalkeeper stands on the goal line at a distance of 30 m from the kick. If the crossbar is 2.44 m high, determine mathematically whether the ball will land inside the goal or fly over the crossbar. [2]

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